



ODEBASICS PRESENTS

GEN AI & DATA SCIENCE BOOTCAMP 3.0

CODEBASICS' STORY



In 2016, **Dhaval Patel**, the founder of Codebasics, created the '**Codebasics**' YouTube channel to share his knowledge as a means to gain mental peace during a significant health struggle. He discovered his passion for teaching and realised his unique ability to simplify complex topics. The love and blessings from thousands of students started to heal his health issues. This transformative experience soon became his life's purpose and even led him to consider leaving his lucrative job at Bloomberg, New York.

In 2020, one of Dhaval's subscribers, **Hemanand Vadivel**, a **data analytics expert** working in Europe, wished to contribute to the Codebasics' community as a 'payback' for what he learned from the Codebasics channel. In 2021, Hemanand began creating impactful videos for Codebasics, and it helped him heal from the trauma of losing family members to Covid-19. Seeing their shared values, Dhaval proposed a collaboration: creating the best, most affordable Power BI course on the planet.



JUNE 2022

After six months of dedicated collaboration, they released the Power BI course in June 2022, which was met with overwhelming positivity. By this time, both Dhaval and Hemanand had already left their high-paying jobs to create 'Codebasics' as a full-time learning tech company. They made this decision after discovering their 'Ikigai', their true purpose: to make a meaningful impact through education, without any gimmicks.

In the era of commoditized education, where millions of courses are offered for profit but lack true value, Dhaval and Hemanand prioritized providing authentic, job-relevant learning experiences with honest pricing. Besides job-related affordable courses, Codebasics offers high-quality free content and conducts several initiatives to support aspirants including mothers returning to work. This embodies their core vision of providing authentic job-related learning experiences to everyone, irrespective of their financial condition.

YOUR TEACHERS ARE INDUSTRY EXPERTS



Dhaval Patel

Data Entrepreneur (13+ Years),
YouTuber,
Ex - Bloomberg, NVIDIA

I have 18 years of experience in Programming and Data Science working for big tech companies like NVIDIA and Bloomberg. I also run a famous YouTube channel called Codebasics where I pursue my passion for teaching.



Hemanand Vadivel

Ex- Data Analytics Manager,
8+ Years in Europe, Microsoft Certified,
Certified Supply Chain Professional

I'm a Mechanical Engineer who transitioned to a full-time Data & Analytics Manager in the UK & Germany. I have delivered 30+ analytics projects over 15+ countries and trained professionals at different levels to equip them with valuable analytics skills.



DATA SCIENCE & AI BOOTCAMP:

WITH PRACTICAL JOB ASSISTANCE + VIRTUAL INTERNSHIP



WHY THIS IS THE MOST EFFECTIVE BOOTCAMP ON THIS PLANET

- ✓ We respect both your time and money. Hence, we recommend you take our short AI & Data Science Survey before enrolling in our Bootcamp.
- ✓ We offer unlimited chat support for doubt clearing through Discord.
- ✓ We provide comprehensive practical job assistance that includes: ATS Resume Builder, Interview Preparation, Mock Interview, Virtual Internships, Building Online Credibility, Job Application and Interview Playbooks.
- ✓ You will experience highly engaging content that provides a cinematic feel, real-world business practice problems, and interactive business
- ✓ In this bootcamp, you won't be working on toy datasets. Instead, you'll tackle complex datasets used by real companies, with millions of records.
- ✓ The projects you will be working on will be related to F & B, Banking, Healthcare, Finance, Hospitality, Automotive, Real Estate & Logistics and more will be added in the future.
- ✓ 100% refund within 30 days and pay the difference feature.

**Scan the
QR Code**
to start the survey!



PYTHON

FOR BEGINNER AND INTERMEDIATE LEARNERS

This project-based learning course will teach Python through hands-on experience with two real-world projects: (1) Data analysis in the hospitality domain and (2) Building an expense tracking system. It is designed for both complete beginners with no prior coding experience and those with some Python knowledge looking to enhance their skills. The course is taught by an industry expert with over 14 years of experience at leading tech companies



6250+
Learners

4.8
Rating

SYLLABUS

- | | |
|---|---|
| <ul style="list-style-type: none">✓ Python Installation – Windows✓ Python Installation – Linux✓ Python Installation – Mac✓ Variables✓ Numbers✓ Strings✓ Lists✓ Install Pycharm✓ If Condition✓ For Loop✓ Functions✓ Dictionary and Tuples | <ul style="list-style-type: none">✓ File Handling✓ Classes and Objects✓ Operator Overloading✓ Inheritance✓ Exception Handling✓ Main Function✓ Numpy Introduction and Benefits✓ Basic Operations✓ Matrix Operations✓ Slicing, Stacking✓ Pandas Introduction and Installation |
|---|---|

- ✓ Modules and Pip
- ✓ Dataframe Basics
- ✓ Read, Write Excel and CSV Files
- ✓ Handle NA values
- ✓ Group By
- ✓ Concat and Merge
- ✓ Data Visualization Using Matplotlib and Seaborn
- ✓ Data Exploration
- ✓ Data Cleaning
- ✓ Data Transformation
- ✓ Insights Generation
- ✓ Set and Frozenset
- ✓ Lists, Dict and Set Comprehensions
- ✓ Dataframe Basics
- ✓ Code Debugging Using PyCharm
- ✓ Working with JSON
- ✓ Generators and Iterators
- ✓ Decorators
- ✓ API
- ✓ Calling APIs with request Package
- ✓ Building APIs with FastAPI
- ✓ Automated Testing with Pytest
- ✓ Data Validation with Pydantic
- ✓ Database CRUD Operations
- ✓ Streamlit Introduction (Frontend)
- ✓ Analytics: Backend (FastAPI) & Frontend (Streamlit)

SQL

BEGINNER TO ADVANCED FOR DATA PROFESSIONALS

Beginners to advanced course for those preparing for a data career (data analyst, data scientist, or data engineer). This course is carefully curated to **simulate real-time organizational** experience to prepare you for the current job market and at the same time provides you with an ultimate learning experience through a **storytelling mode** that you would see



7940+
Learners

4.9
Rating

SYLLABUS

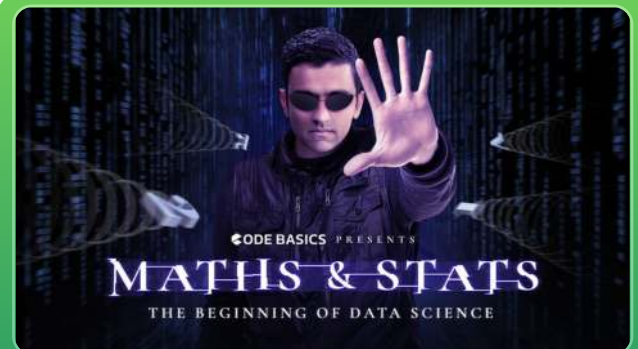
- ✓ Install MySQL: Windows
- ✓ Install MySQL: Linux, Mac
- ✓ Retrieve Data Using Text Query (SELECT, WHERE, DISTINCT, LIKE)
- ✓ Retrieve Data Using Numeric Query (BETWEEN, IN, ORDER BY, LIMIT, OFFSET)
- ✓ Summary Analytics (MIN, MAX, AVG, GROUP BY)
- ✓ HAVING Clause
- ✓ Calculated Columns (IF, CASE, YEAR, CURYEAR)
- ✓ Joins (INNER, LEFT, RIGHT, FULL, CROSS)
- ✓ Join More Than Two Tables
- ✓ Subqueries
- ✓ ANY, ALL Operators
- ✓ Correlated Subquery
- ✓ Common Table Expression (CTE)
- ✓ CTE Benefits & Other Applications
- ✓ Database Normalization and Data Integrity
- ✓ Data Types: Numeric (INT, DECIMAL, FLOAT, DOUBLE)
- ✓ Data Types: String (VARCHAR, CHAR, ENUM)
- ✓ Data Types: Date, Time (DATETIME, DATE, TIME, YEAR, TIMESTAMP)
- ✓ Data Types: JSON, Spatial (JSON, GEOMETRY)

- ✓ Primary Key
 - ✓ Foreign Key
 - ✓ Entity Relationship Diagram - ERD
 - ✓ INSERT, UPDATE, DELETE Statement
 - ✓ ETL, Data Warehouse
 - ✓ OLAP vs OLTP, Data Catalog
 - ✓ Fact vs Dimension Table
 - ✓ Star vs Snowflake Schema
 - ✓ Data Import
 - ✓ Simplified: What is Kanban?
- ✓ User-Defined Functions
 - ✓ Stored Procedures
 - ✓ Database Views
 - ✓ Window Functions OVER Clause, ROW_NUMBER, RANK, DENSE_RANK
 - ✓ Database Triggers
 - ✓ Database Events
 - ✓ User Accounts and Privileges
 - ✓ Database Indexes

MATH AND STATISTICS

FOR AI, DATA SCIENCE

Learn the key concepts of Math and Statistics that lay the foundations for a strong data science career. This course is carefully curated to simulate real-time organizational experience to prepare you for the current job market and at the same time provides you with an ultimate learning experience through storytelling and intuitive



1360+
Learners

5.0
Rating

SYLLABUS

- ✓ Types of Data
- ✓ Pie Chart and Bar Chart
- ✓ Histograms and Line Chart
- ✓ Scatter and Bubble Plot
- ✓ Univariate vs. Bivariate vs. Multivariate Analysis
- ✓ Numpy Introduction and Benefits
- ✓ Basic Operations
- ✓ Matrix Operations
- ✓ Slicing, Stacking
- ✓ Descriptive vs. Inferential Statistics
- ✓ Measures of Central Tendency: Mean, Median, Mode
- ✓ Percentile
- ✓ Measures of Dispersion: Range, IQR
- ✓ Box or Whisker Plot
- ✓ Outlier Treatment Using IQR and Box Plot
- ✓ Measures of Dispersion: Variance and Standard Deviation
- ✓ Correlation vs Causation
- ✓ Probability Basics
- ✓ Addition and Multiplication Rule
- ✓ Conditional Probability and Bayes Theorem
- ✓ Distribution

- ✓ Skewness
- ✓ Normal Distribution
- ✓ Detect Outliers Using Normal Distribution
- ✓ Z Score
- ✓ Standard Normal Distribution (SND)
- ✓ Kanban Project Management
- ✓ Random Sampling & Sample Bias
- ✓ The Law of Large Numbers
- ✓ Central Limit Theorem, Sampling Distribution
- ✓ Standard Error
- ✓ Confidence Interval
- ✓ Null vs Alternate Hypothesis
- ✓ Z Test, Rejection Region
- ✓ p-Value
- ✓ Statistical Power & Effect Size
- ✓ A/B Testing
- ✓ A/B Testing Using Z Test
- ✓ T-test
- ✓ Chi-squared Distribution
- ✓ Chi-squared Test of Goodness of Fit
- ✓ Chi-squared Test of Independence

MASTER MACHINE LEARNING FOR DATA SCIENCE & AI

BEGINNER TO ADVANCED

Learn Machine Learning from an AI entrepreneur with extensive industry experience and a popular YouTube channel (Codebasics) with 1 million subscribers. This course takes you from beginner to advanced levels, providing deep intuition on algorithms, engaging cinematic experiences, end-to-end projects, and hands-on coding practice. Designed for easy understanding, even for high school students, all at an affordable price.



2210+
Learners

5.0
Rating

SYLLABUS

- ✓ Introduction to Machine Learning
- ✓ Classification vs Regression
- ✓ Supervised vs Unsupervised Learning
- ✓ Simple Linear Regression
- ✓ Multiple Linear Regression
- ✓ Cost Function
- ✓ Derivatives and Partial Derivatives
- ✓ Chain Rule
- ✓ Gradient Descent
- ✓ Model Evaluation
- ✓ Data Preprocessing: One Hot Encoding
- ✓ Polynomial Regression
- ✓ Overfitting and Underfitting
- ✓ L1 and L2 Regularization
- ✓ Bias Variance Trade Off
- ✓ Introduction to Classification
- ✓ Logistic Regression: Binary Classification
- ✓ Model Evaluation: Accuracy, Precision and Recall, F1 Score, Confusion Matrix
- ✓ Logistic Regression: Multiclass Classification
- ✓ Cost Function: Log Loss
- ✓ Support Vector Machine (SVM)
- ✓ Data Pre-processing: Scaling
- ✓ Sklearn Pipeline
- ✓ Naive Bayes

- ✓ Decision Tree
- ✓ Handle Class Imbalance
- ✓ Introduction to Ensemble Learning
- ✓ Majority Voting, Average and Weighted Average
- ✓ Bagging
- ✓ Random Forest
- ✓ Boosting: AdaBoost
- ✓ Gradient Boosting
- ✓ XGBoost
- ✓ Model Evaluation: ROC Curve & AUC
- ✓ Cost Benefit Analysis Using ROC in Sklearn
- ✓ K Fold Cross Validation
- ✓ Stratified K Fold Cross Validation
- ✓ Hyperparameter Tuning
- ✓ 10 Stages of AI Project Lifecycle
- ✓ Requirements and Scope of Work (SOW)
- ✓ Data Collection, Data Cleaning & Exploratory Data Analysis
- ✓ Feature Engineering
- ✓ Model Selection & Training
- ✓ Model Fine Tuning
- ✓ Model Deployment
- ✓ Monitoring and Feedback Using ML Ops
- ✓ Variance Inflation Factor (VIF)
- ✓ K Means Clustering

DEEP LEARNING

BEGINNER TO ADVANCED

Deep Learning: Beginner to Advanced is a comprehensive course that covers everything from the fundamentals of neural networks to advanced architectures such as CNNs, RNNs and transformers. Taught by an experienced AI professional, the course emphasizes practical learning through hands-on exercises and real-world projects in PyTorch. By the end, you'll have the skills to confidently build, train, and deploy deep learning models for real-world applications.



240+
Learners

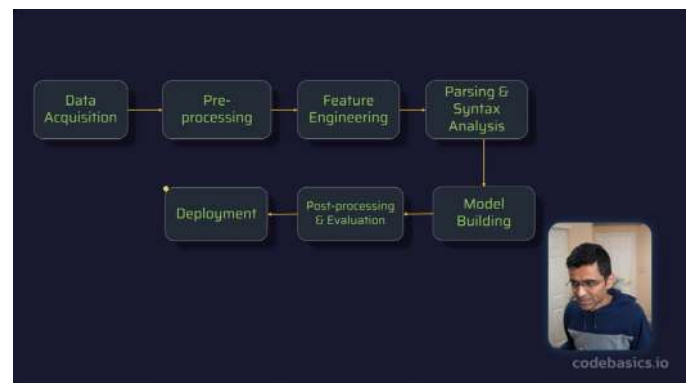
SYLLABUS

- ✓ Introduction to Deep Learning
- ✓ Neural Networks
- ✓ Deep Learning vs Statistical ML
- ✓ Neural Network Architectures
- ✓ Applications of Deep Learning
- ✓ PyTorch vs TensorFlow
- ✓ GPU, TPU
- ✓ Neuron, Perceptron and MLP
- ✓ Activation Function
- ✓ Sigmoid, ReLU, Tanh, SoftMax
- ✓ PyTorch Installation
- ✓ PyTorch Tensor Basics
- ✓ Autograd in PyTorch
- ✓ Training through Backpropagation
- ✓ Gradient Descent
- ✓ Theoretical Foundation, PyTorch Implementation
- ✓ Batch GD vs Mini Batch GD vs SGD
- ✓ Handwritten Digits Classification
- ✓ Model Optimization
- ✓ Gradient Descent with Momentum
- ✓ Adam Optimizer
- ✓ Regularization
- ✓ Dropout Regularization

- ✓ Batch Normalization
- ✓ Hyperparameter Tuning
- ✓ Optuna
- ✓ CNN
- ✓ CIFAR10 Image Classification using CNN
- ✓ Data Augmentation
- ✓ Transfer Learning
- ✓ Pre-trained Models: ResNet, EfficientNet, MobileNet
- ✓ RNN
- ✓ Vanishing Gradient Problem
- ✓ LSTM
- ✓ Transformer Architecture
- ✓ Word Embeddings
- ✓ Attention Mechanism
- ✓ Hugging Face: BERT Basics
- ✓ Model Training with CNN
- ✓ Model Training using Transfer Learning
- ✓ Streamlit App
- ✓ FastAPI Server

NATURAL LANGUAGE PROCESSING (NLP)

This course takes you through the core concepts of NLP, covering topics such as text preprocessing, representation techniques, and advanced tools like Hugging Face and BERT. With a focus on hands-on exercises, it provides practical exposure to key NLP workflows, ensuring you gain the skills required for NLP-driven roles in today's job market.

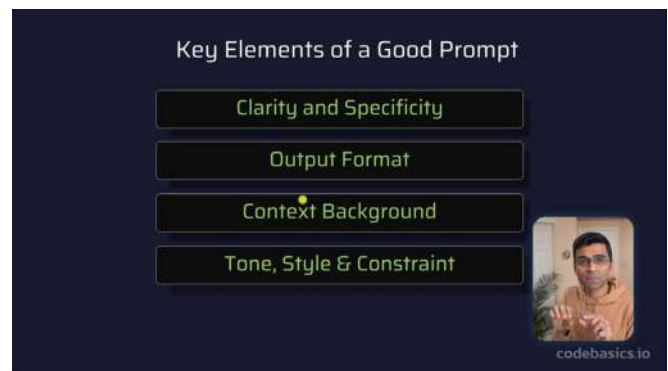
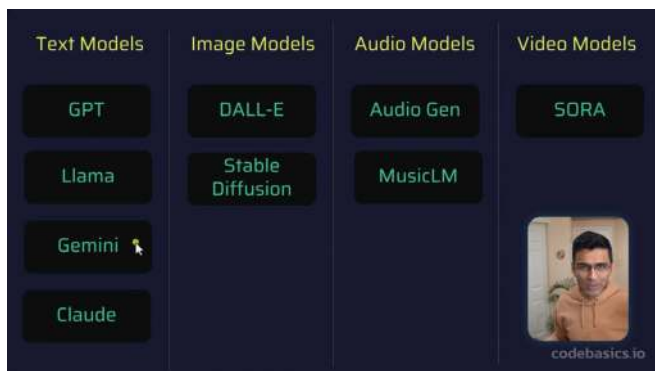


SYLLABUS

- ✓ Introduction to NLP
- ✓ NLP Pipeline
- ✓ Tools Overview
- ✓ Tokenization Techniques
- ✓ Stemming & Lemmatization
- ✓ Part of Speech (POS) Tagging
- ✓ Stop Words
- ✓ Named Entity Recognition (NER)
- ✓ Regular Expressions (Regex)
- ✓ Bag of Words, n-grams, TF-IDF
- ✓ Word Embeddings
- ✓ Word Embeddings
- ✓ News Classification with Spacy
- ✓ Introduction to Hugging Face
- ✓ Pipelines and Tokenizers
- ✓ BERT and Model Fine Tuning

GEN AI & AGENTIC AI

This course explores the core concepts of Generative AI and Agentic AI, covering large language models, RAG, vector databases, prompt engineering, MCP and multi-agent system design. You'll work on 4 real-world projects: including a real estate assistant using RAG, an e-commerce chatbot with database integration, Customer Support AI Agent and a business project built using agentic workflows. Through hands-on projects using LangChain, LangGraph, Amazon AgentCore, Claude MCP SDK and CrewAI, you'll learn to design and evaluate autonomous AI agents, covering advanced topics like agent architecture, multi-agent collaboration, MCP, and fine-tuning for real-world applications.



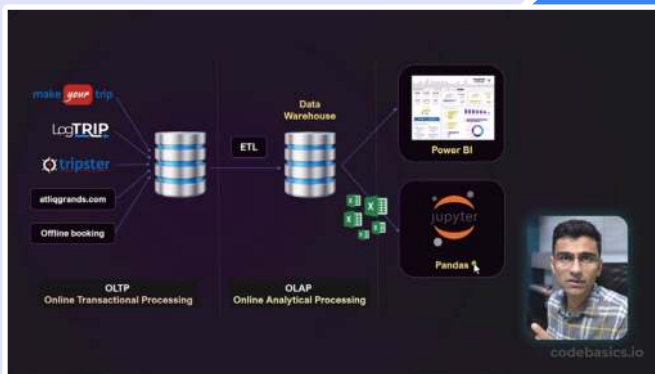
SYLLABUS

- ✓ Introduction to Gen AI & Agentic AI
- ✓ Application of Gen AI & Agentic AI
- ✓ Large Language Models
- ✓ AI Agent with Custom tools
- ✓ Context Window
- ✓ Temperature
- ✓ Model Context Protocol (MCP)
- ✓ Agentic AI Evaluation
- ✓ Gen AI Application Development Steps
- ✓ Hallucinations, Security, Cost
- ✓ Vector Database
- ✓ RAG (Retrieval Augmented Generation)
- ✓ Streamlit UI Development
- ✓ Multi Agent Systems
- ✓ Zero-Shot, One-Shot, Few-Shot Prompting
- ✓ LangChain Installation and Setup
- ✓ Calling LLM from LangChain
- ✓ Prompt Templates & Chains
- ✓ Chromadb
- ✓ Metadata Filtering
- ✓ Fine Tuning an LLM
- ✓ SQLite Database Integration
- ✓ LangGraph Crash course
- ✓ CrewAI crash course
- ✓ Amazon Bedrock AgentCore

BUSINESS PROJECTS

01 HOSPITALITY DOMAIN DATA ANALYSIS

PYTHON



```
Out[96]: booking_id property_id booking_date check_in_date checkout_date no_guests room_category booking_platform ratings_0

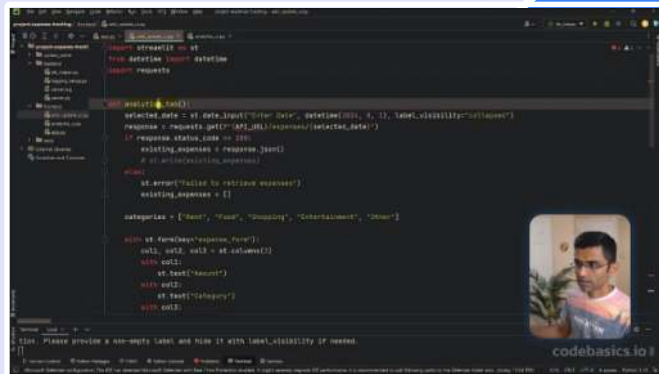
In [97]: df_bookings_all.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 134573 entries, 0 to 134572
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   booking_id            134573 non-null object
1   property_id           134573 non-null int64
2   booking_date           134573 non-null object
3   check_in_date          134573 non-null object
4   checkout_date          134573 non-null object
5   no_guests              134573 non-null float64
6   room_category          134573 non-null object
7   booking_platform       134573 non-null object
8   ratings_given          56676 non-null float64
9   booking_status         134573 non-null object
10  revenue_generated      134573 non-null int64
11  revenue_realized       134573 non-null int64
12  property_name          134573 non-null object
13  category               134573 non-null object
14  city                   134573 non-null object
dtypes: float64(2), int64(3), object(10)
memory usage: 16.4+ MB
```

By completing this project, you will:

- ✓ Gain deep domain knowledge in the hotel industry, specifically in data analysis for operations, customer behaviour, revenue management, and market trends.
- ✓ Acquire essential technical skills using Python's Pandas library, including data cleaning, transformation, EDA, statistical analysis, and visualization.
- ✓ Recognize the high relevance of data-driven insights in today's business world for making informed decisions, optimizing operations, and enhancing customer experiences, not just in the hotel industry but across various sectors.

02 EXPENSE TRACKING SYSTEM — PYTHON



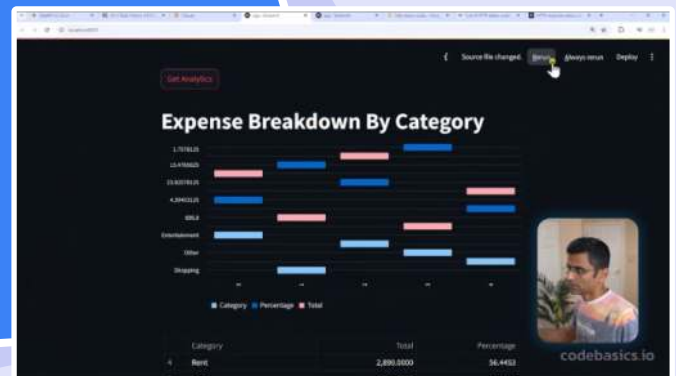
```
import streamlit as st
from datetime import datetime
import requests

def add_expense():
    selected_date = st.date_input("Select Date", datetime(2024, 4, 1), label_visibility="collapsed")
    response = requests.get(f"{API_URL}/expenses/{selected_date}")
    if response.status_code == 200:
        existing_expenses = response.json()
        st.write(existing_expenses)
    else:
        st.error("Failed to retrieve expenses")
        existing_expenses = []

    categories = ["Rent", "Food", "Shopping", "Entertainment", "Other"]

    with st.form(key="expense_form"):
        col1, col2, col3 = st.columns(3)
        with col1:
            st.text_input("Amount")
        with col2:
            st.text_input("Category")
        submit = st.form_submit_button("Add")

    if submit:
        # Add logic to add expense to database
        pass
```



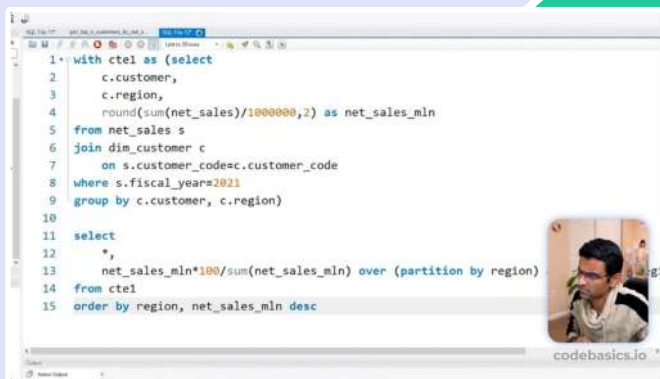
By completing this project, you will:

- ✓ Develop practical knowledge of expense management systems by building an application that tracks and analyses day-to-day expenses, such as rent, shopping, and food, providing users with a comprehensive view of their spending habits.
- ✓ Master key technical skills, including Python basics, database interactions with MySQL, backend development using FastAPI, and frontend design using the Streamlit framework, gaining end-to-end application development experience.
- ✓ Understand the value of analytics in financial decision-making by implementing features like category-wise and month-wise expense breakdowns, making this project relevant for aspiring software engineers, data scientists, and data analysts.

03

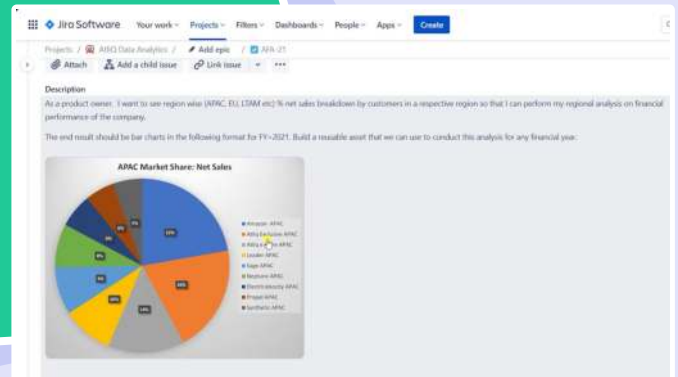
FINANCE AND TOP N INSIGHTS

SQL



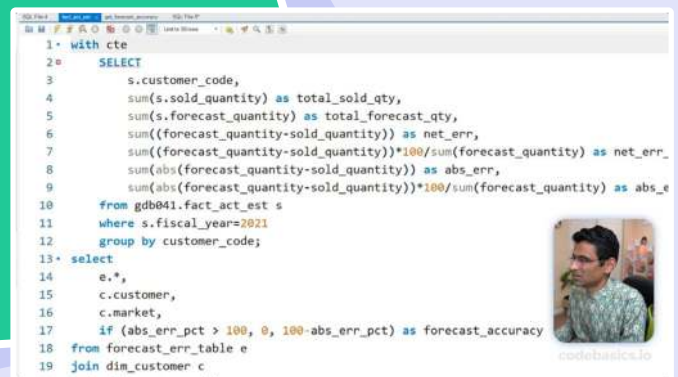
```
1 with cte1 as (select
2   c.customer,
3   c.region,
4   round(sum(net_sales)/1000000,2) as net_sales_mln
5 from net_sales s
6 join dim_customer c
7   on s.customer_code=c.customer_code
8 where s.fiscal_year=2021
9 group by c.customer, c.region)
10
11 select
12   *,
13   net_sales_mln*100/sum(net_sales_mln) over (partition by region)
14 from cte1
15 order by region, net_sales_mln desc
```

A screenshot of a SQL query in a code editor. The query uses a CTE to calculate net sales in millions for each customer and region in 2021, then uses a window function to calculate the percentage of net sales for each customer within their region. The results are ordered by region and then by net sales in descending order.



- ✓ Gain domain knowledge in finance analytics for the consumer goods industry, focusing on generating reports for key metrics.
- ✓ Develop technical skills in efficient SQL querying, utilizing User-Defined Functions, creating and executing stored procedures, and leveraging views for data analysis.
- ✓ Understand the significance of data-driven financial decision-making in today's competitive business landscape.
- ✓ Drive sustainable growth by utilizing automated reporting solutions, such as stored procedures, to provide accurate and real-time information for financial analysis and decision-making.

04 SUPPLY CHAIN ANALYTICS AND MODEL OPTIMISATION

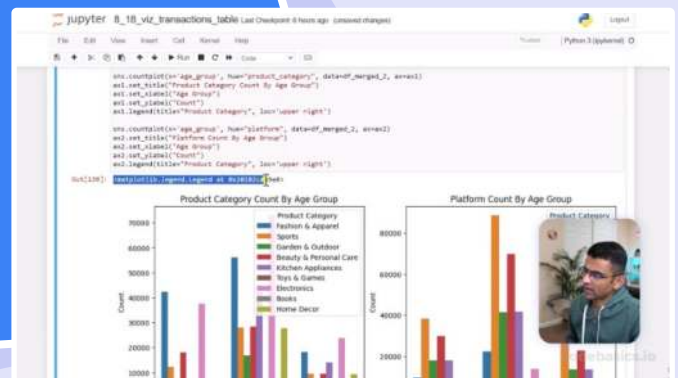
SQL

By completing this project, you will:

- ✓ Gain domain knowledge in supply chain analytics for the consumer goods industry, focusing on forecasting errors and absolute error calculation.
- ✓ Develop technical skills in creating helper tables, managing triggers, and optimizing query execution time.
- ✓ Understand the significance of accurate demand forecasting for consumer goods companies and how supply chain analytics can achieve it.
- ✓ Acquire practical experience through hands-on exercises and real-world examples in consumer goods supply chain analytics.

05

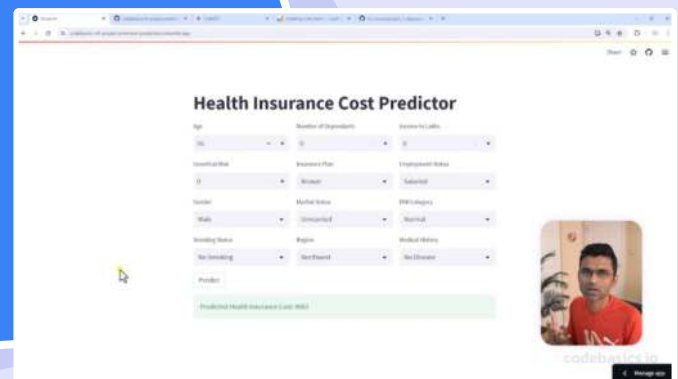
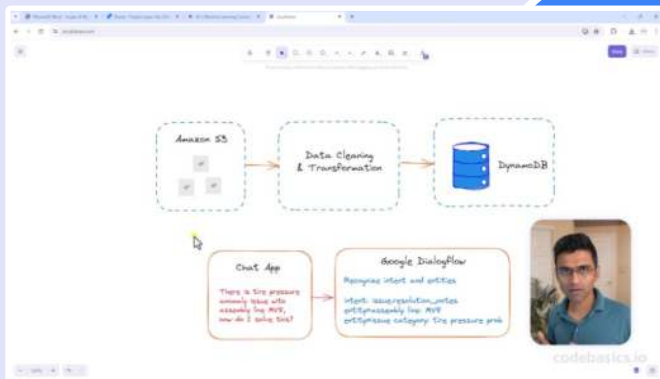
ATLIQO BANK PROJECT

MATH AND
STATISTICS**By completing this project, you will:**

- ✓ Gain hands-on experience in the banking sector by working on a large-scale data analysis project aimed at identifying the target market for a new credit card launch, simulating real-life industry challenges with over 50,000 records.
- ✓ Master essential data analysis techniques, including data validation, cleaning, handling outliers, MySQL setup, and data visualization using Python, while performing critical tasks like credit score analysis, correlation studies, and hypothesis testing.
- ✓ Understand the role of A/B testing and statistical decision-making in marketing campaigns, providing insights into customer segmentation and campaign effectiveness, which are applicable across various industries.

06 HEALTHCARE PREMIUM PREDICTION (REGRESSION)

MACHINE LEARNING

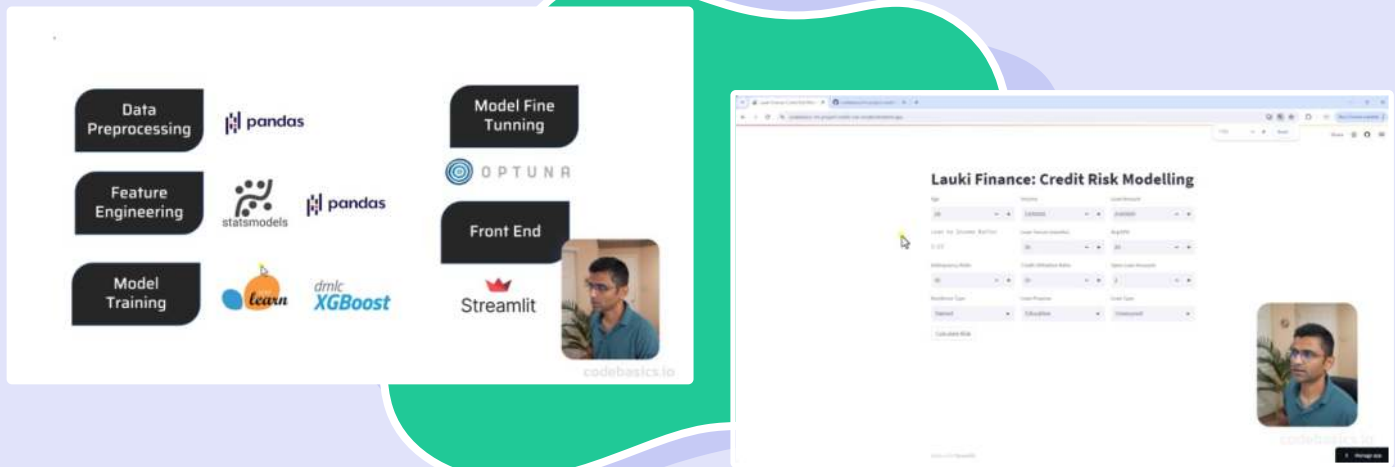


By completing this project, you will:

- ✓ Gain domain knowledge in the healthcare industry by predicting insurance premiums using machine learning regression models, covering real-world tasks like data collection, model training, and deployment.
- ✓ Develop advanced technical skills in data cleaning, exploratory data analysis (EDA), feature engineering, and regression model fine-tuning, achieving high accuracy while performing error analysis and segmentation.
- ✓ Understand the practical application of machine learning in predicting financial outcomes, and learning to build and deploy models using Python, Streamlit, and other tools for real-time decision-making across industries.

07 CREDIT RISK MODELLING (CLASSIFICATION)

MACHINE LEARNING



By completing this project, you will:

- ✓ Gain valuable insights into the financial domain, particularly non-banking financial companies (NBFCs) and credit approval processes, by developing a credit risk model to classify potential defaulters.
- ✓ Acquire technical expertise in data cleaning, exploratory data analysis (EDA), feature engineering (WOE/IV), and model training, with a focus on classification techniques, model fine-tuning using Optuna, and evaluation metrics like KS Statistic and Gini Coefficient.
- ✓ Learn how to deploy machine learning models by building a Streamlit app, delivering business presentations, and applying credit risk modelling principles across various financial sectors.

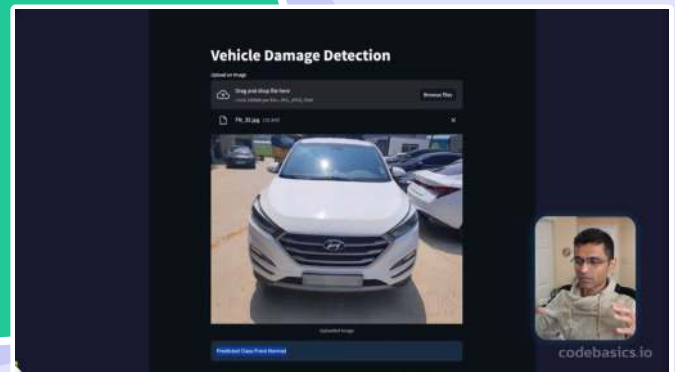
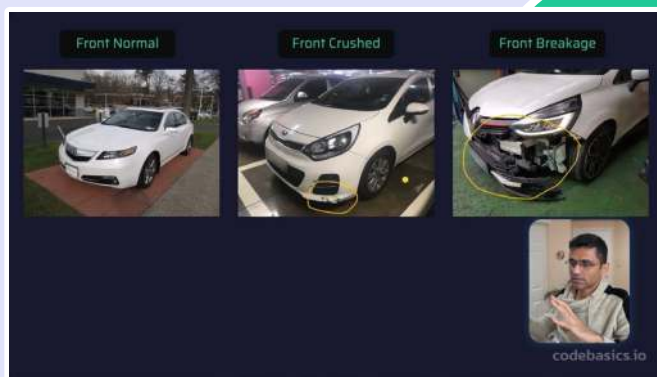
08 BEVERAGE PRICE RANGE PREDICTION

The screenshot shows a web application titled "CodeX Beverage: Price Prediction". It features a form with 16 input fields arranged in a 4x4 grid. The fields are: Age (30), Gender (M), Zone (Metro), Occupation (Entrepreneur), Income Level (INR) (15L - 25L), Consumption Frequency (weekly) (5-7 times), Current Brand (Established), Preferable Consumption Size (Medium (500 ml)), Awareness of other Brands (2 to 4), Reasons for choosing brands (Quality), Taste Preference (Traditional), Purchase Channel (Online), Packaging Preference (Premium), Health Concerns (Medium (Moderately health-conscious)), and Typical Consumption Situations (Active (eg. Sports, gym)). Below the form is a "Calculate Price Range" button and a text box displaying "Price Range: 200-250 INR".

By completing this project, you will:

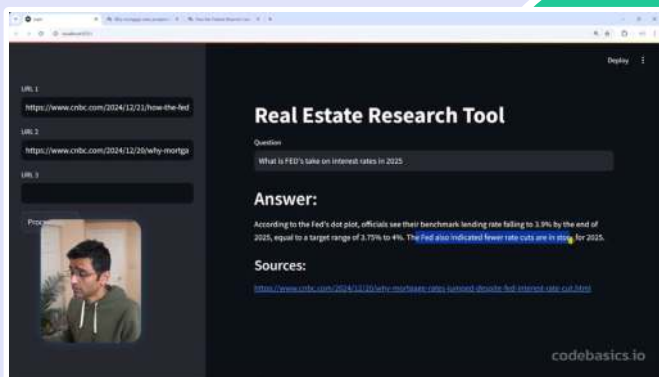
- ✓ Gain valuable insights into market pricing strategies by building a predictive model that forecasts optimal price ranges for beverages, helping businesses avoid overselling or underselling based on survey data.
- ✓ Develop essential technical skills in machine learning, including data cleaning, validation, feature engineering, and model deployment using Python, ensuring the creation of an accurate and reliable classification model.
- ✓ Understand how predictive models drive business growth and competitiveness by enabling data-driven pricing decisions, applicable across various sectors beyond the beverage industry.

09 CAR DAMAGE DETECTION USING CNN



- ✓ Gain domain knowledge in the automotive industry by developing a deep learning model to detect car damages using Convolutional Neural Networks (CNN) and Transfer Learning, covering real-world tasks like data preprocessing, model training, and deployment.
- ✓ Acquire advanced technical skills in data augmentation, fine-tuning CNN models, hyperparameter optimization using Optuna, and error analysis to improve model accuracy and performance.
- ✓ Learn how to deploy deep learning models using Python, Streamlit, and FastAPI, creating a real-time car damage detection system that can be applied in the automotive industry for automated damage assessment.

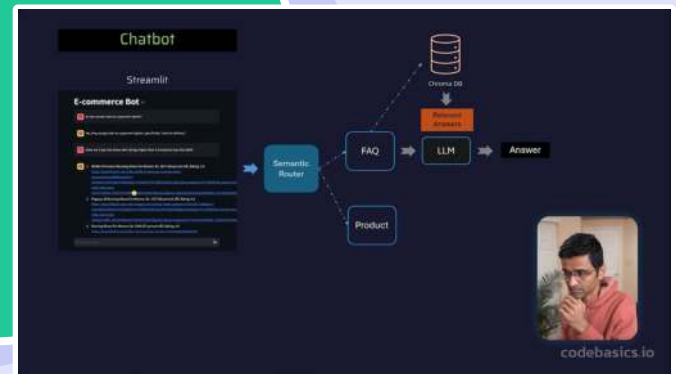
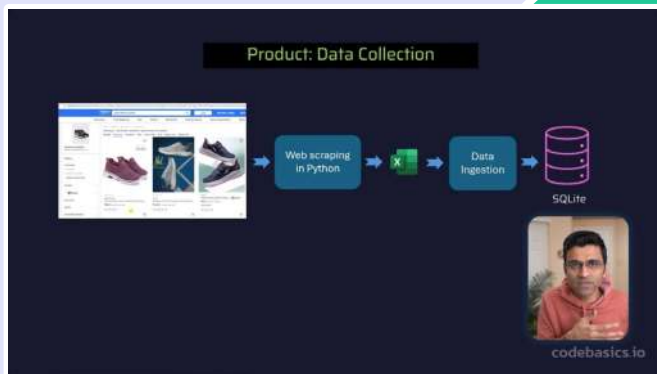
10 REAL ESTATE ASSISTANT USING RAG



- ✓ Gain domain knowledge in the real estate industry by developing a generative AI solution using Retrieval-Augmented Generation (RAG) to assist with property-related queries, applying advanced techniques for efficient data retrieval and answer generation.
- ✓ Develop technical expertise in integrating RAG with Chroma DB for data storage, implementing retrieval and answer generation mechanisms, and fine-tuning the system for accurate and relevant responses.
- ✓ Learn to deploy the solution with a user-friendly Streamlit interface, enabling seamless interaction for end-users and gaining experience in building scalable, interactive AI systems.

11

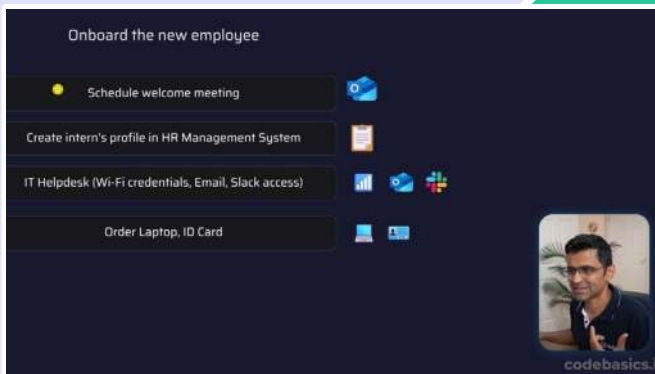
E-COMMERCE CHATBOT



- ✓ Gain domain knowledge in the e-commerce industry by building an AI-powered chatbot capable of handling FAQs, product queries, and assisting customers with purchase decisions, utilizing advanced NLP techniques and semantic routing.
- ✓ Acquire technical expertise in integrating an SQLite database for dynamic product information handling, as well as implementing a chatbot to provide real-time responses and personalized customer assistance.
- ✓ Learn to deploy the chatbot using a Streamlit UI for seamless interactions, gaining hands-on experience in developing real-time customer-facing AI applications.

12

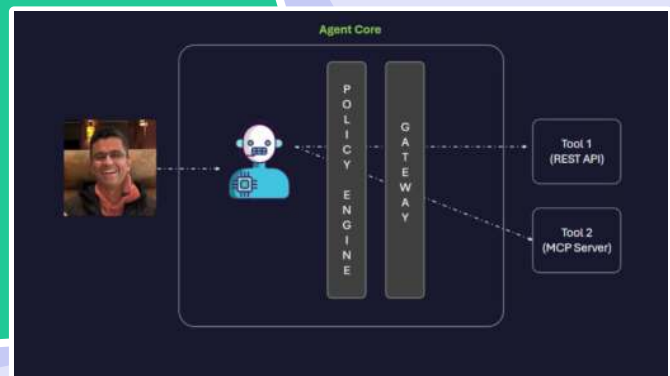
AGENTIC AI PROJECT



- ✓ Tackle a real-world HR challenge by developing an autonomous AI agent that streamlines and automates the employee onboarding process, reducing manual HR workload and enhancing process efficiency.
- ✓ Master technical skills to transform existing HR APIs into Model Context Protocol (MCP) servers, enabling seamless integration and interoperability for advanced agentic applications.
- ✓ Gain hands-on experience for connecting MCP servers with large language models like Claude, unlocking powerful automation and intelligent workflow capabilities in HR operations.

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CUSTOMER CARE AI AGENT



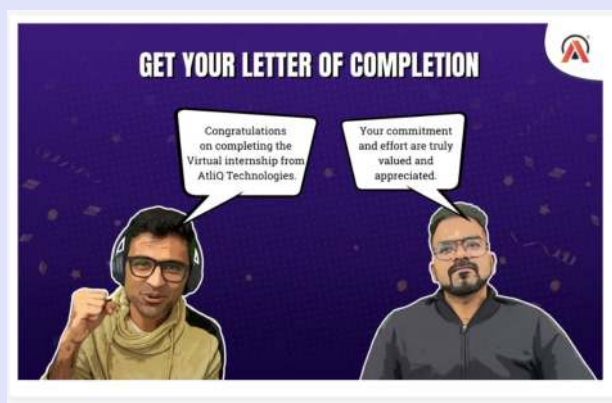
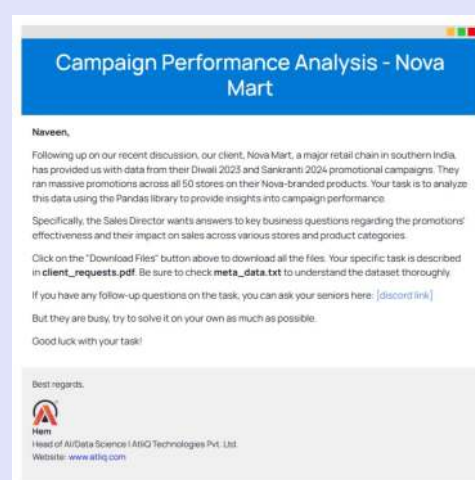
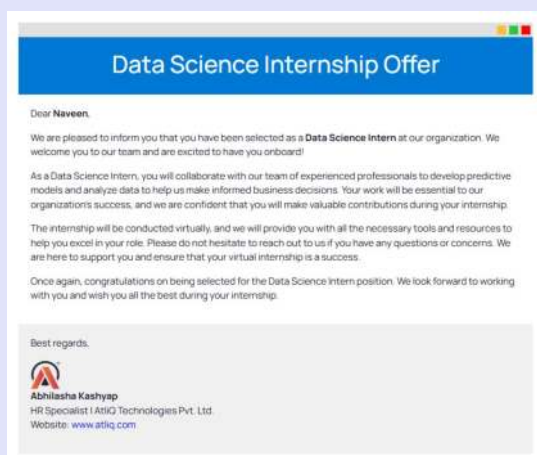
- ✓ Build a production-ready AI agent using Amazon Bedrock AgentCore, covering the full lifecycle from local development to secure deployment, scaling, and runtime execution on AWS.
- ✓ Add long-term and short-term memory to the AI Agent using Agentcore Memory and push it to production.
- ✓ Learn how enterprise AI agents are operated in practice, including observability, security, identity, memory and moving beyond demos to real deployable systems.

VIRTUAL INTERNSHIPS

In our virtual Internship, you'll gain practical experience in Data Science. You'll gain good knowledge in data cleaning, data validation, reporting, feature engineering, model deployment and client handling. You will also learn scrum project management and improve debugging skills. With a problem-solving mindset, you'll be ready to tackle any data challenge and make a meaningful impact.

OPTIONAL

In our Virtual Internship 2, you will work on client projects focused on AI solutions. For Stale Fruit Detection, you will develop a deep learning model to automate freshness checks in cold storage warehouses. In the RAG-Based Q&A System for Healthcare Professionals, you will build a pipeline to query PubMed articles and generate relevant responses using a Streamlit UI.



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STALE FRUIT DETECTOR FOR A LOGISTICS COMPANY

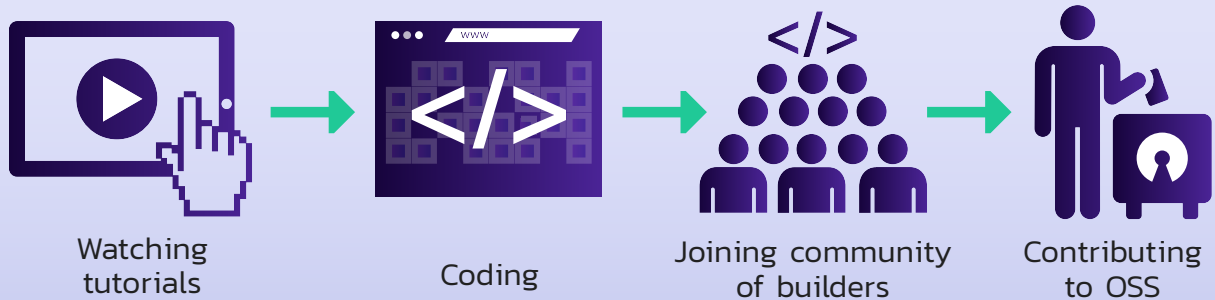
- ✓ Gain domain knowledge in the agri-tech industry by developing a CNN-based deep learning model to detect stale fruit in cold storage warehouses, applying real-world techniques such as data augmentation, model training, and fine-tuning.
- ✓ Develop advanced technical skills in transfer learning, data preprocessing, and model optimization to enhance detection accuracy, ensuring reliable classification of fresh and stale fruit in real-world
- ✓ Learn how to deploy a user-friendly Streamlit app that allows clients to upload images for real-time stale fruit detection, improving warehouse operations and quality control through automation.

15

RAG BASED Q & A SYSTEM FOR HEALTHCARE PROFESSIONALS

- ✓ Gain domain knowledge in the healthcare industry by developing a Q&A system for healthcare professionals, leveraging Retrieval-Augmented Generation (RAG) to provide accurate, context-driven answers from PubMed articles.
- ✓ Develop advanced technical skills in data ingestion, document retrieval, and RAG-based question answering using large language models (LLM) like Llama 3, integrating a robust pipeline for efficient querying and response generation.
- ✓ Learn to deploy the solution with a Streamlit UI, enabling real-time interactions for healthcare professionals to query articles and receive relevant responses in an easy-to-use interface.

BUILD IN PUBLIC MENTORSHIP



- ✓ Build in Public Mentorship Program that helps you transition from **consumers of knowledge to real-world builders** by focusing on hands-on product development, opensource contributions, and engineering excellence.
- ✓ Gain mentorship to work on real OSS and product-grade projects through weekly jamming sessions contributing to frameworks, popular open-source projects, fellowship side projects, and guided paper-to-implementation workflows.
- ✓ Develop builder credibility and career leverage by learning how to contribute meaningfully to open source, conduct independent research, and build a strong personal brand across GitHub, LinkedIn, blogs, and developer communities.

LIVE PROBLEM-SOLVING SESSION

The Live Gen AI & Data Science Problem-Solving Sessions help you learn how experts design, debug, and deploy **REAL-WORLD AI SYSTEM** – live, end-to-end, and in real time.

In each session, experienced AI practitioners walk you through **ACTUAL GEN AI AND DATA SCIENCE PROBLEMS** from production environments such as building RAG systems, designing AI agents, optimizing ML pipelines, or evaluating LLM outputs and show you exactly how they solve them step by step.

YOU'LL GET:

- ✓ Live expert walkthroughs of real AI & ML problems: Watch how industry practitioners' reason through open-ended AI problems, trade-offs, failures, and final design decisions.
- ✓ Case studies from companies building AI at scale: Learn how teams at companies like Uber, Netflix, Airbnb, fintechs, and AI startups build, evaluate, and deploy Gen AI and ML systems in production.
- ✓ End-to-end system thinking: From problem framing and data understanding to model selection, prompt design, evaluation, deployment, and monitoring.
- ✓ Hands-on, immediately applicable insights: Patterns, architectures, and best practices you can directly apply to your own Gen AI projects and research work.
- ✓ Two live sessions every month with AI practitioners, stay current with fast-moving AI tools, frameworks, and real-world use cases
- ✓ One-year access to all live AI & Data Science sessions from the date of your enrollment.

PRACTICE ROOM

The Practice Room helps you build real Gen AI and Data Science skills through hands-on, real world challenges.

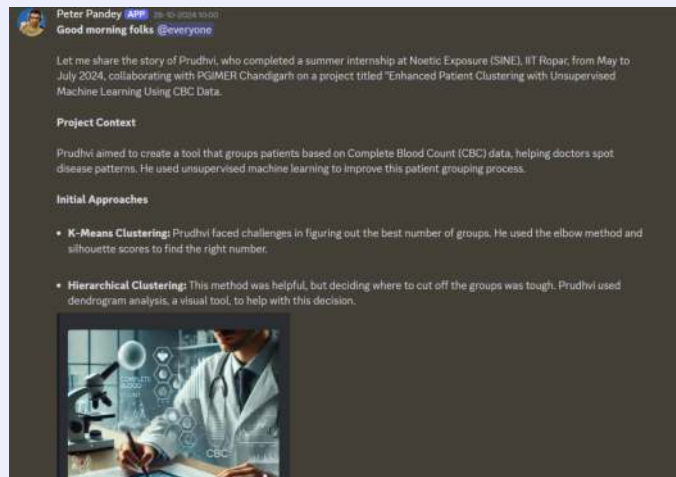
Each room – Python, SQL, ML, DL, LLMs, RAG, Agentic AI, ML pipelines, gives you a realistic environment to solve the kinds of problems AI and data professionals face, step by step.

YOU'LL GET:

- ✓ Work on real scenarios like data preprocessing, visualization, model training, prompt engineering, RAG pipelines, agent orchestration, evaluation, etc.
- ✓ Validate your code, pipelines, and workflows to ensure your solutions actually work as intended.
- ✓ Short, focused challenges based on real use cases – Each exercise is concise, goaloriented, and inspired by real-world AI and data problems.
- ✓ By the end, you'll know how to apply your skills the way AI Engineers and Data Scientists do in real projects.

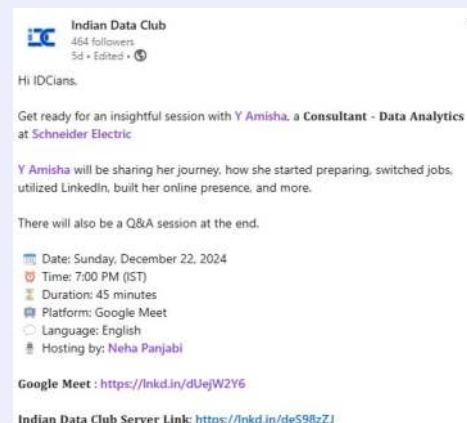
DAILY DISCORD CHAT SUPPORT

Get unlimited doubt-clearing support through our active Discord community. This dedicated platform allows you to interact with fellow learners and mentors, ensuring your questions are answered promptly and fostering collaborative learning throughout your journey.



LEARNING COMMUNITY INDIAN DATA CLUB (IDC)

Join our vibrant new community, the Indian Data Club (IDC), designed to enhance your learning experience. Participate in regular meetups, engage in weekly quizzes, and attend live sessions to stay connected, learn collaboratively, and grow with like-minded individuals.

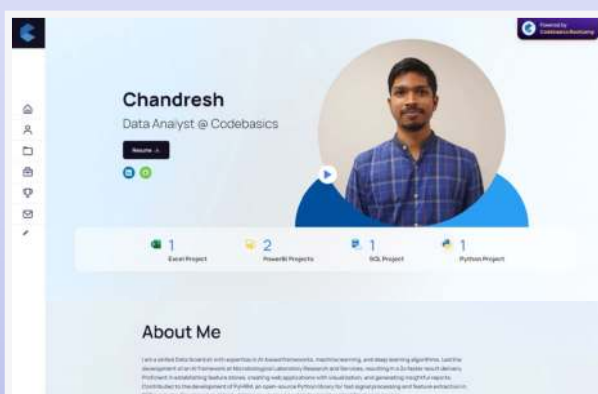
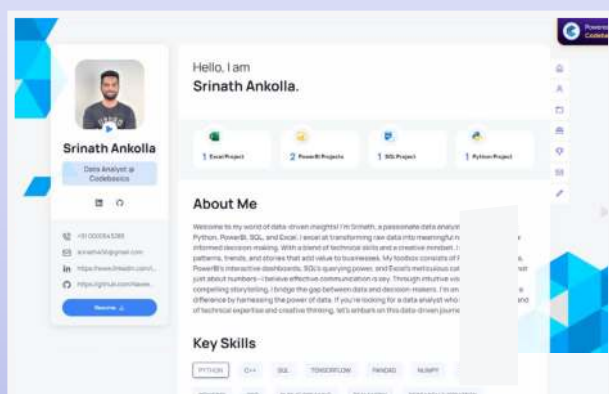
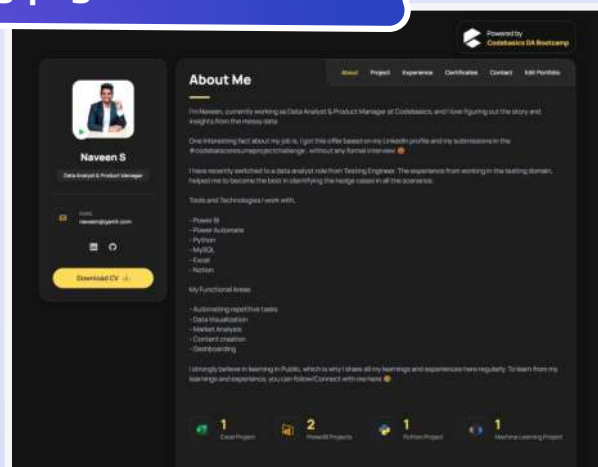
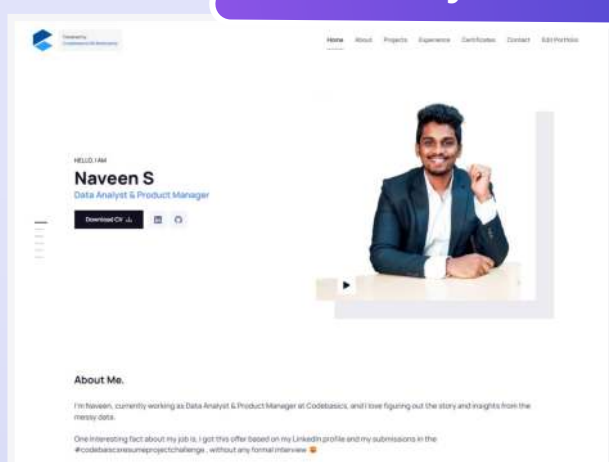


PROJECT PORTFOLIO WEBSITE

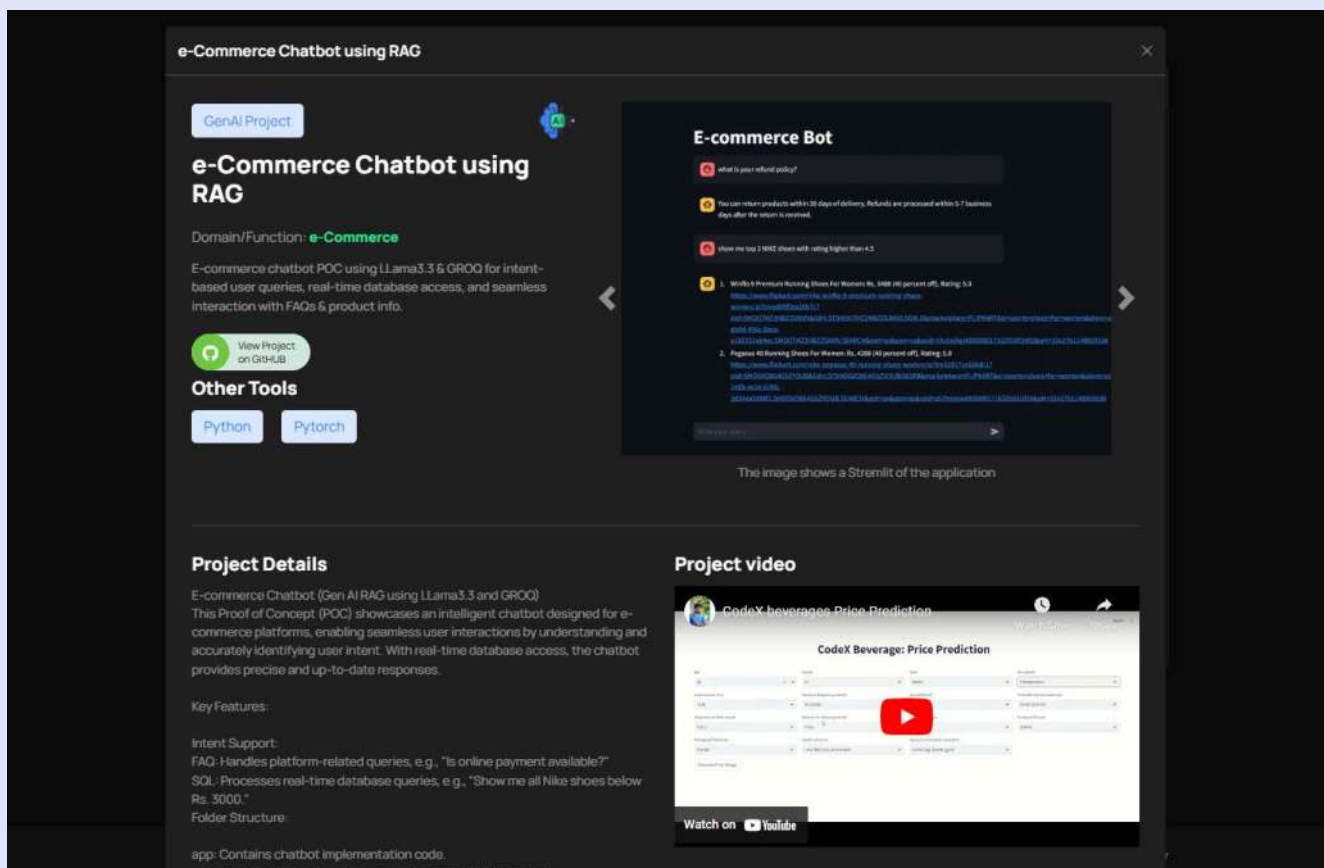
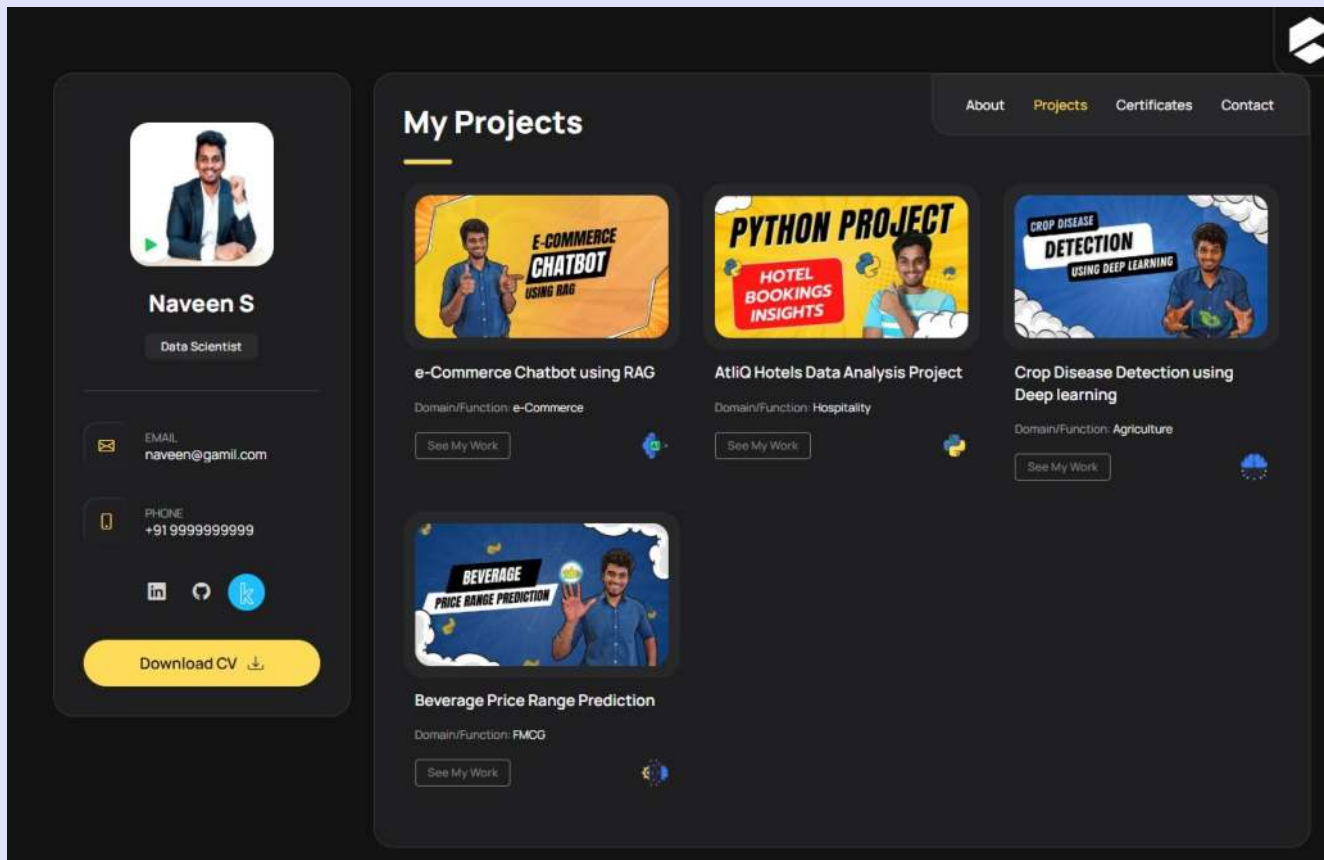
Having a **PROJECT PORTFOLIO WEBSITE** is a fantastic way to stand out from the crowd and to show the proof of your work. This is the right way to convince the hiring managers to call you for an interview especially if you are a fresher or someone from a non-technical background. We advised our learners to create a portfolio website but noticed that they found it challenging to create a compelling portfolio website from scratch due to the lack of proper tools in the

Hence, we ended up creating a feature to automate the **PORTFOLIO WEBSITE** within our bootcamp so our learners can focus on practice and interview preparation. To meet various preferences and needs, we now offer different themes for portfolio websites. Choose the style that best represents your professional image and effectively showcases your work.

This is how your landing page would look like

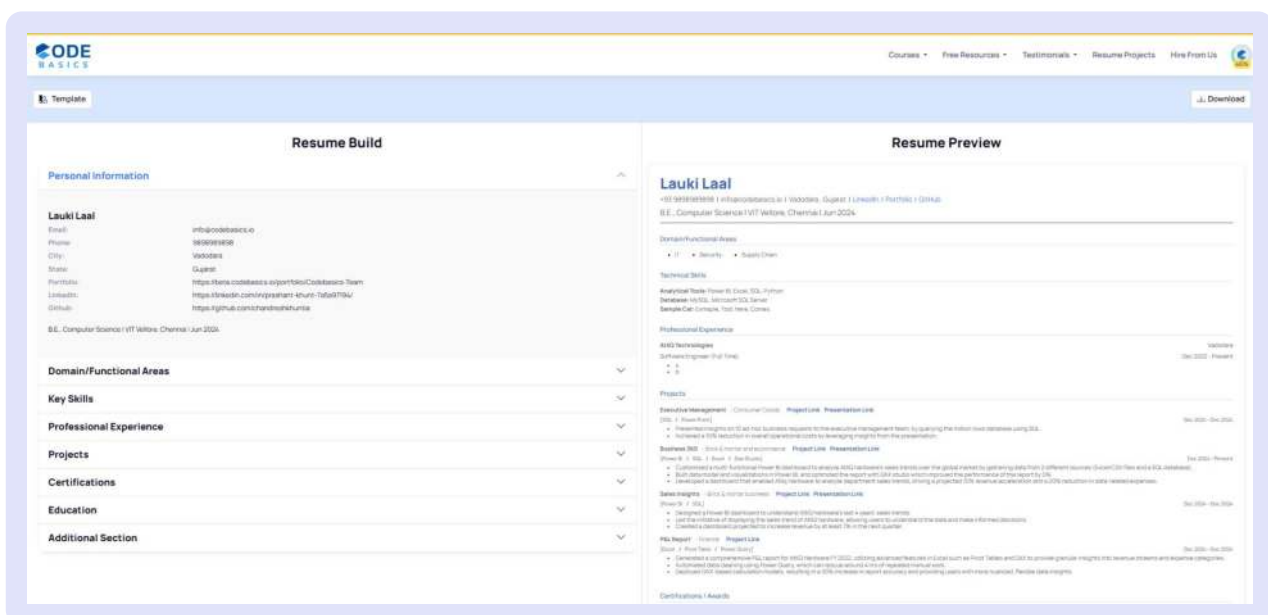
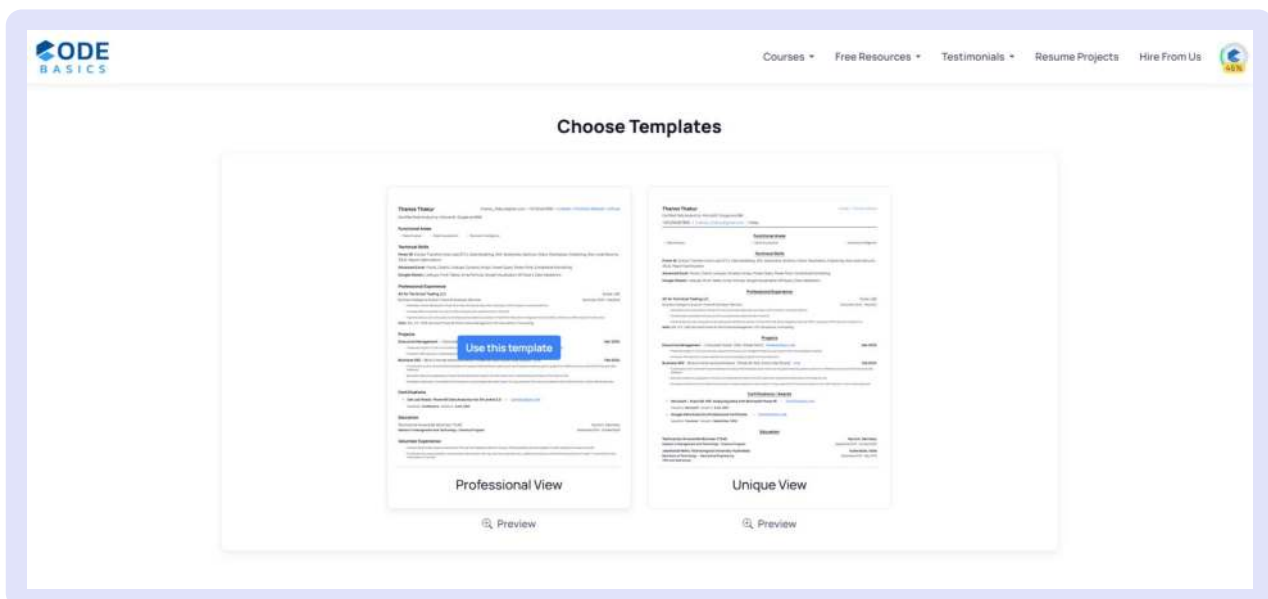


More templates available in our Bootcamp



ATS RESUME BUILDER

Our ATS Resume Builder helps you create professional resumes effortlessly. Enter your details, and the tool formats them in a structured way. It ensures your resume looks clear and well-organized. Save time and focus on applying for jobs. Perfect for showcasing your skills effectively.



ONLINE CREDIBILITY

This module is all about using the internet to boost your chances of finding a job, getting better career opportunities, and freelancing. It will also teach you how to build your own personal brand for long-term benefits

By the end of this module, you will have an updated LinkedIn profile picture, banner image, and about section. Additionally, you will receive the following action steps.

- 01 3-step framework** to build your online credibility with proven results.
- 02 5 Advanced Tips for super growth**
- 03 10 post-writing templates** for high engagement
- 04 Curated list of people** to Follow / Connect for Data Professionals

Step 1 Get your LinkedIn Profile Right



Peter Pandey
Data Scientist

Talks about #datascience
#deeplearning #machinelearning
and #python



Step 2 Engage Meaningfully



Tony Sharma
Senior Data Scientist

You are replying to Tony Sharma's post



I concur with your suggestions and would like to add a few more that would be beneficial for individuals new to the field of data science.

Step 3 Create Your Own Posts



Peter Pandey
Data Scientist

3mo • 🌐

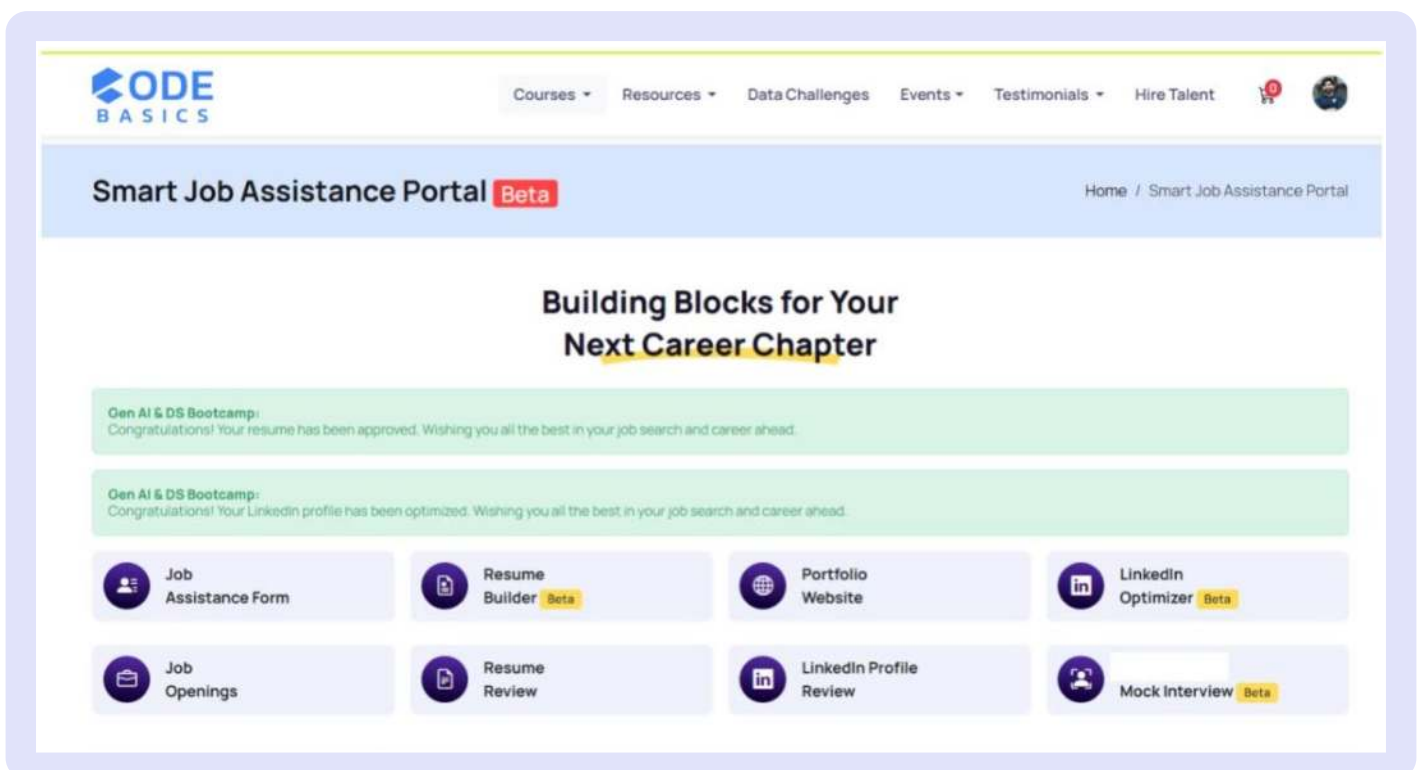
Hello everyone,
In this post, I would like to share with all of you about my approach, mindset, and the steps I took during the execution of the project.

SMART JOB ASSISTANCE PORTAL

Through this job portal, you can review and optimize your resume, project portfolio, and LinkedIn profile with the assistance of a dedicated talent management team.

You can also find relevant job openings and receive comments from the talent management team, such as which keywords to use and which projects to discuss. Additionally, there is space to add and track applications.

This portal will continuously help you upgrade to ensure we assist you in the best possible way.

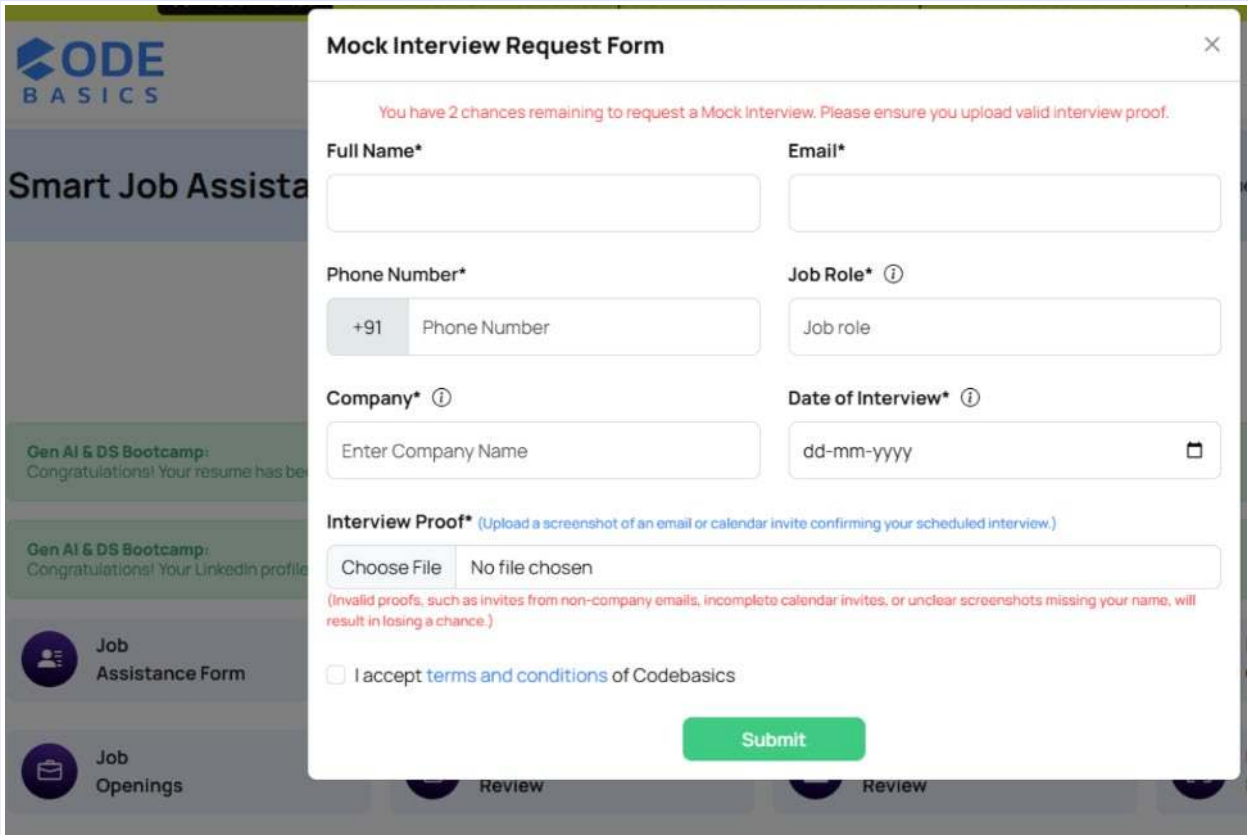


MOCK INTERVIEW

Get two personalized 1-on-1 mock interview sessions upon confirming your actual interview schedule.

Receive tailored feedback to refine your responses and boost your confidence.

Practice common and role-specific questions with industry experts to prepare effectively for real interviews. Build confidence and ace your next interview with ease.



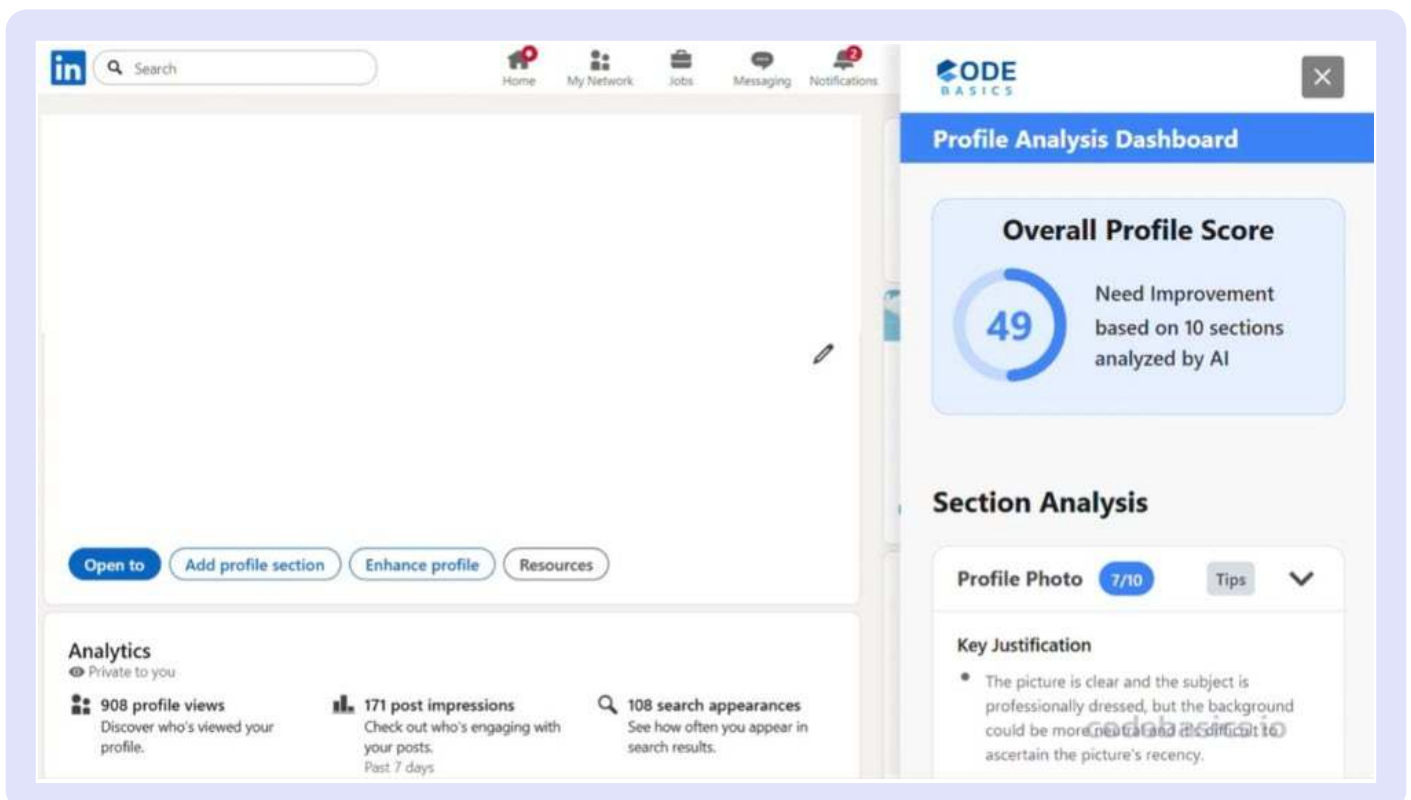
The screenshot shows a 'Mock Interview Request Form' overlaying a blurred background of the Codebasics website. The form is titled 'Mock Interview Request Form' and includes a close button (X) in the top right corner. A red warning message at the top states: 'You have 2 chances remaining to request a Mock Interview. Please ensure you upload valid interview proof.' The form contains several input fields: 'Full Name*' and 'Email*' (text inputs), 'Phone Number*' (a dropdown for '+91' followed by a text input for the number), 'Job Role*' (a text input with an information icon), 'Company*' (a text input with an information icon), and 'Date of Interview*' (a text input with a calendar icon and an information icon). Below these fields is the 'Interview Proof*' section, which includes a file upload button 'Choose File' and a status 'No file chosen'. A red warning message below the upload section states: '(Invalid proofs, such as invites from non-company emails, incomplete calendar invites, or unclear screenshots missing your name, will result in losing a chance.)'. At the bottom of the form, there is a checkbox for 'I accept terms and conditions of Codebasics' and a green 'Submit' button. The background website shows the 'ODE BASICS' logo, 'Smart Job Assistant', and sections for 'Gen AI & DS Bootcamp' and 'Job Assistance Form'.

LINKEDIN OPTIMIZER

LinkedIn Optimizer with AI that can review your profile and suggest corrections to make your profile noticeable to recruiters.

AI-driven profile scoring & gap analysis - Get an objective score for your LinkedIn profile with section-wise feedback that highlights what's working and what's holding you back from visibility.

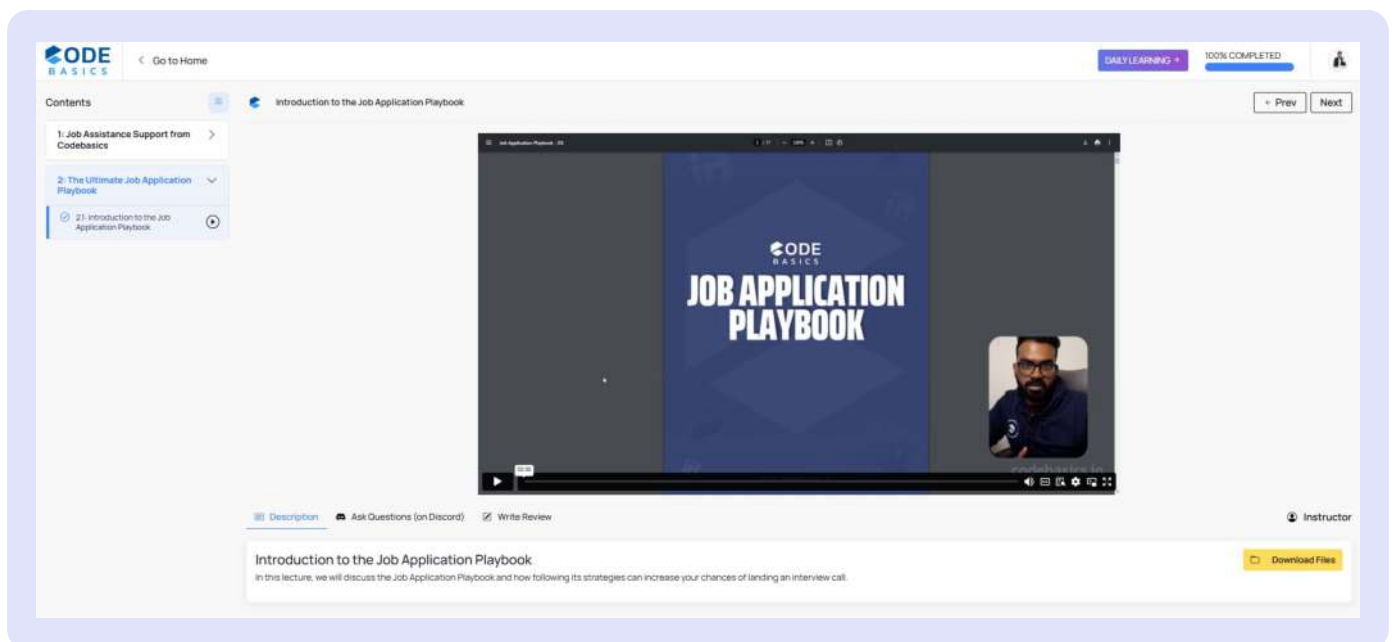
Actionable, section-specific improvement suggestions - Receive clear recommendations for your headline, summary, experience, and skills focused on clarity, relevance, and recruiter searchability (not generic advice).



JOB APPLICATION PLAYBOOK

Spending just 20–30 minutes with this playbook can save you at least 30 days of effort in your journey.

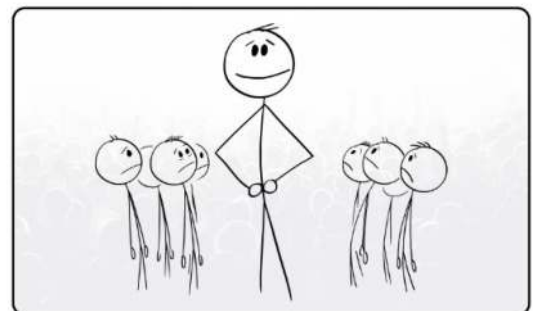
Crack the game of applying for jobs with our Playbook! From comprehensive steps to actionable tips, learn strategies that truly work and boost your chances of landing an interview call.



Time Investment:



If you invest just **30 minutes** to go through this playbook, you can save up to 15 days on your job hunt by streamlining your applications and improving your chances of getting noticed.

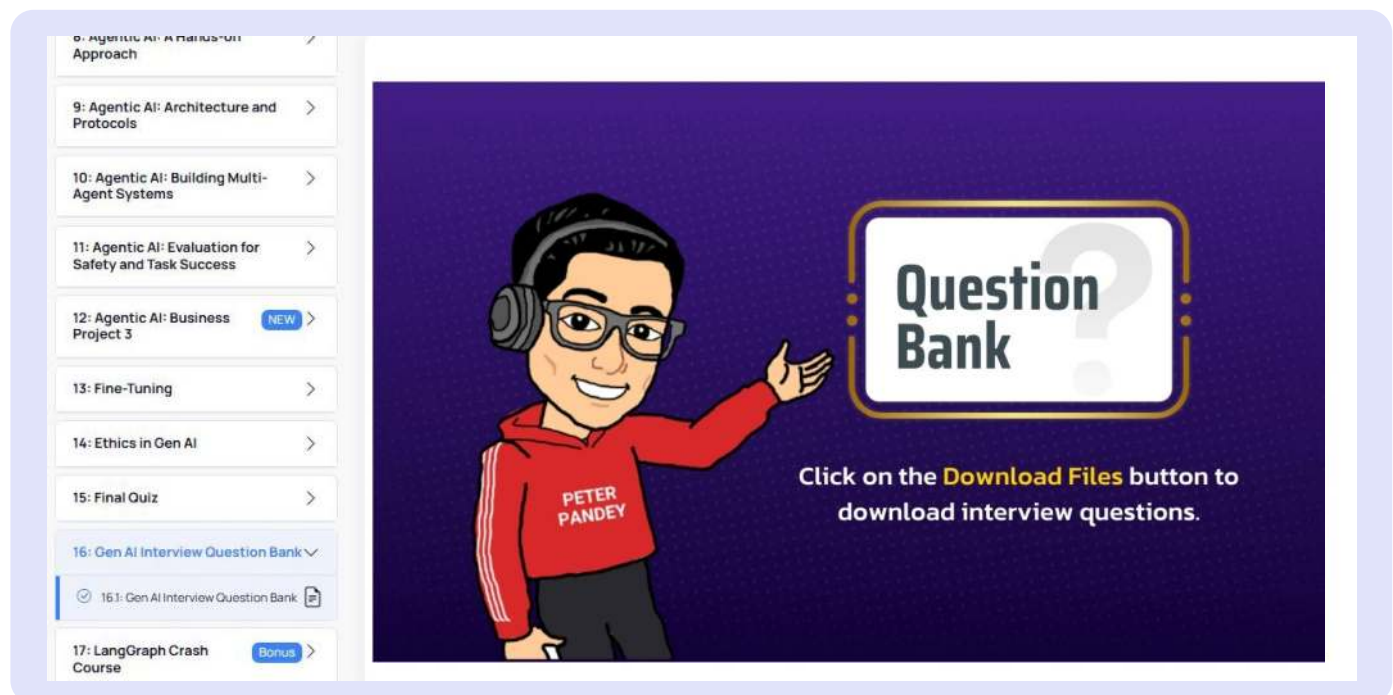


Whether you're starting out or advancing your career, this playbook will help you get noticed by employers and land your dream job.

Standing out in today's job market takes more than just effort—it's about presenting yourself effectively, and this guide shows you how.

INTERVIEW QUESTIONS

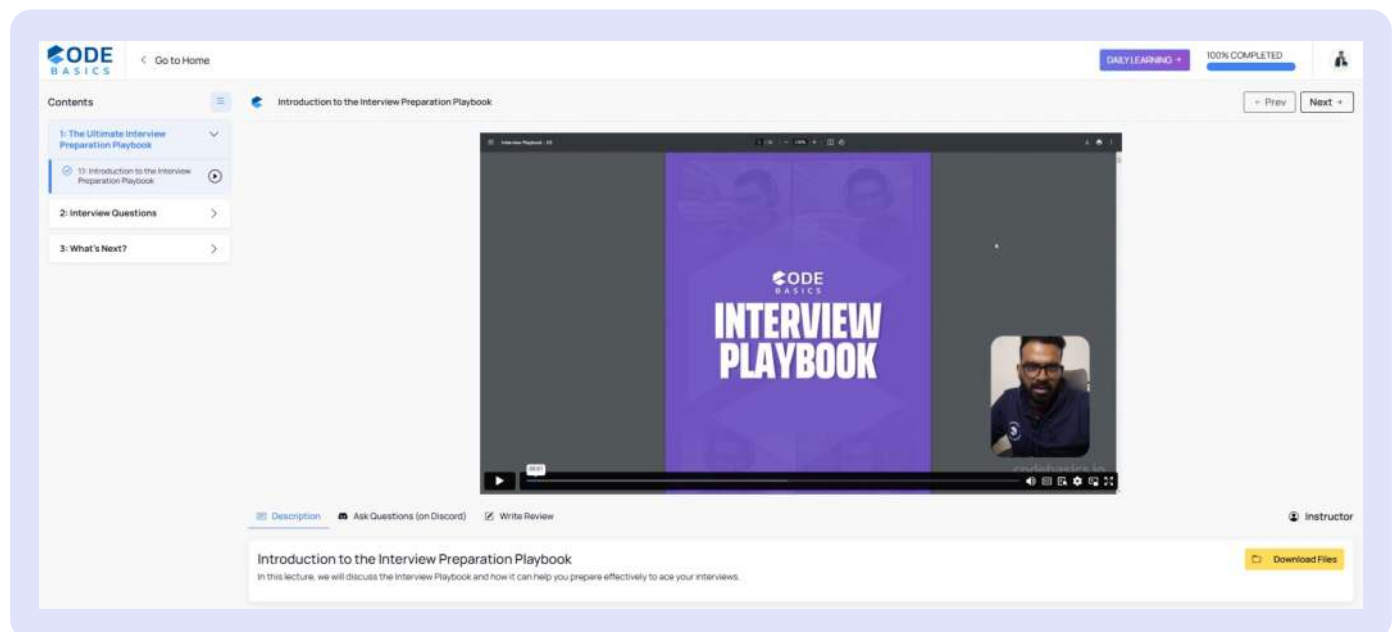
Prepare smarter with our **Interview Question Bank**, featuring module-wise questions across **Python, Math & Stats, Machine Learning, NLP, Deep Learning, Gen AI and Agentic AI**. Practice real-world technical and conceptual questions, strengthen your problem-solving approach, and build the confidence to excel in interviews for AI and Data roles.



The screenshot displays the CodeBasics website interface. On the left, a sidebar lists various modules, including '8: Agentic AI: A Hands-on Approach', '9: Agentic AI: Architecture and Protocols', '10: Agentic AI: Building Multi-Agent Systems', '11: Agentic AI: Evaluation for Safety and Task Success', '12: Agentic AI: Business Project 3' (marked as 'NEW'), '13: Fine-Tuning', '14: Ethics in Gen AI', '15: Final Quiz', '16: Gen AI Interview Question Bank' (highlighted), '16.1: Gen AI Interview Question Bank' (with a document icon), and '17: LangGraph Crash Course' (marked as 'Bonus'). The main content area features a dark purple background with a cartoon character of Peter Pan, wearing a red hoodie with 'PETER PANDEY' written on it and headphones, pointing towards a 'Question Bank' box. The box contains the text 'Question Bank' and a large question mark. Below the box, a text prompt reads: 'Click on the **Download Files** button to download interview questions.'

INTERVIEW PLAYBOOK

Ace your interviews with our comprehensive playbook! Follow a step-by-step guide to prepare for interviews, present your best self, clarify questions with the interviewer, and confidently handle technical queries. Learn how to showcase your skills effectively and leave a lasting positive impression. Your ultimate guide to mastering interviews with confidence.



FROM FOUNDER'S DESK

Hello, I'm **Dhaval Patel**, and I want to share how and why I started the Codebasics YouTube channel. It all began during one of the darkest periods of my life. In **2011**, while working for Bloomberg, I was diagnosed with **Ulcerative colitis**, a painful autoimmune disease. Despite trying numerous treatments, my health continued to decline. By **2015**, I felt hopeless and **constantly in**



During this time, I stumbled upon a life-changing quote:

"Life is 10% what happens to you and 90% how you react."

Determined to shift my focus and find joy, I sought an activity that I enjoyed and that allowed me to help others. **Given my teaching abilities and inclination to share knowledge, I started the Codebasics YouTube channel.** Teaching became an outlet for me, enabling me to make a positive impact and share my expertise.

With each "Thank you" comment I received on my videos, my sense of purpose and will to live grew stronger. Over time, the Codebasics community flourished, and we now have over **1.4 Million+ subscribers** on YouTube and more than **44K+ paid learners**. I've left my full-time job in the year 2022 to dedicate myself to Codebasics fully. My goal is to continue sharing my knowledge and creating a **positive impact** through this platform.

Use my story as to remind yourself that life is not always going to be easy but you can rise above that and make the world a better place to live for yourself and others.



150K+



50K+



1.4M+



635K+



info@codebasics.io



www.codebasics.io

December - 2025